$y = \sum_{n=0}^{\infty} a_n x^n$

Sorun 10

$$\begin{bmatrix} \frac{1}{2} & -1 \\ 1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Sözdeş-

$$\begin{vmatrix} 1/2 - \lambda & -1 \\ 1 & 1/2 - \lambda \end{vmatrix} = \frac{1}{4} - \frac{1}{2}\lambda - \frac{1}{2}\lambda + \lambda^2 + 1 = 0$$

$$\lambda^2 - \lambda + \frac{5}{4} = 0$$

$$\Delta = b^2 - 4ac = 1 - 4 \cdot 1 \cdot \frac{5}{4} = -4$$

$$\frac{-b \pm \sqrt{\Delta}}{2a} = \frac{1 \pm 2i}{2} = \frac{1}{2} + i \qquad \frac{1}{2} - i$$

Özvektör $\lambda_1 = \frac{1}{2} + i$

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = 0$$

$-i\alpha - \beta = 0 \quad \beta = \beta$

$\alpha - i\beta = 0 \quad \alpha = i\beta$

$$\begin{bmatrix} i\beta \\ \beta \end{bmatrix} \cdot e^{(\frac{1}{2}+i)t}$$

$$= \beta \begin{bmatrix} i \\ 1 \end{bmatrix} e^{\frac{1}{2}t} \cdot e^{it}$$

$$= \beta \begin{bmatrix} i \\ 1 \end{bmatrix} e^{\frac{1}{2}t} \cdot e^{it}$$

↳ Bir özvektör bulmak yeterli

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) e^{\frac{1}{2}t} \cdot e^{it}$$

$$= \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) e^{\frac{1}{2}t} \cdot (\cos t + i \sin t)$$

$$= \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{\frac{1}{2}t} \cos t - \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{\frac{1}{2}t} \sin t \right) + i \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{\frac{1}{2}t} \sin t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{\frac{1}{2}t} \cos t \right)$$

$$\begin{bmatrix} x_1(t) \\ y_1(t) \end{bmatrix} = \begin{bmatrix} \sin t \\ \cos t \end{bmatrix} e^{\frac{1}{2}t}$$

$$\begin{bmatrix} x_2(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix}$$

$L[F] = \frac{5}{s}$ $\quad F = 5$

$y(0) = 0 \quad y'(0) = 0$

$L[y''] + 2L[y'] + 10L[y] = \frac{5}{s}$

$s^2L[y] - s\,y(0) - y'(0) + 2(sL[y] - y(0)) + 10L[y] = \frac{5}{s}$

$L[y](s^2 + 2s + 10) = \frac{5}{s}$

$$L[y] = \frac{5}{s(s^2+2s+10)} = \frac{A}{s} + \frac{Bs+C}{(s^2+2s+10)}$$

$$5 = A(s^2 + 2s + 10) + Bs^2 + Cs$$

$A + B = 0 \qquad 2A + C = 0$

$10A = 5$

$A = \frac{1}{2} \quad B = -\frac{1}{2} \quad C = -1$

$$L[y] = \frac{1}{2}\cdot\frac{1}{s} - \frac{1}{2}\left(\frac{1s+2}{s^2+2s+10}\right)$$

$$= \frac{1}{2} - \frac{1}{2}\left(e^{-t}\cos(3t) + \frac{e^{-t}\sin 3t}{3}\right)$$

$$= \frac{1}{2} - \frac{e^{-t}\cos(3t)}{2} - \frac{e^{-t}\sin(3t)}{6}$$

$$\frac{s+1}{(s+1)^2+3^2} + \frac{1}{(s+1)^2+3^2}$$

Terslaplace

$e^{-t}\cdot\cos(3t) + \frac{1}{3}\cdot e^{-t}\cdot\sin 3t$

$x' = 2x$

$y' = -x + 4y$

$z' = -3x + 6y + 2z$

$$\begin{bmatrix} 2 & 0 & 0 \\ -1 & 4 & 0 \\ -3 & 6 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 2-\lambda & 0 & 0 \\ -1 & 4-\lambda & 0 \\ -3 & 6 & 2-\lambda \end{bmatrix}$$

$\lambda = 4$

$$\Rightarrow \begin{bmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ -3 & 6 & -2 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = 0$$

$-2\alpha = 0 \Rightarrow \alpha = 0$ $\quad \alpha = \alpha$

$-1\beta = 0 \Rightarrow \beta = 0$ $\quad \beta = 0$

α cinsinden yazalım $\quad -3\alpha + 6\beta = 2\gamma$

$$\gamma = -\frac{3\alpha}{2}$$

$$\begin{bmatrix} \alpha \\ 0 \\ -\frac{3\alpha}{2} \end{bmatrix} e^{4t}$$

β ya göre

$\alpha = 0$

$\beta = \beta$

$\gamma = 3\beta$

$$\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} e^{4t}$$

İlk önce denklemi y'' türevini yalnız bırakarak formal hale getirilir.

$$y'' + \underbrace{\frac{(x^2+1)}{2x}}_{p} y' + \underbrace{\frac{\ln(1-x)}{2x}}_{q} y = 0$$

$r(r-1) + p_0 r + q_0 = 0$ denklemi yazılır ve kök bulunur

burda $p_0 = \lim_{x \to 0} x\,p(x)$ ve $q_0 = \lim_{x \to 0} x^2 q(x)$

$$p_0 = \lim_{x \to 0} \frac{(x^2+1)}{2} = \frac{1}{2}$$

$$q_0 = \lim_{x \to 0} x \frac{\ln(1-x)}{2} = 0$$

o zaman karakteristik denklemeniz

$$r(r-1) + \frac{1}{2} r + 0 = 0$$

$$r^2 - \frac{1}{2} r = 0$$

$$r\left(r - \frac{1}{2}\right) = 0$$

$$r_1 = 0 \qquad r_1 = \frac{1}{2}$$

Şimdi köklerin farkına bakılır

$r_1 - r_2 = \frac{1}{2}$ (tam sayı değil)

o zaman

$$y_1(x) = x^{r_1} \sum_{n=0}^{\infty} c_n x^n \,, \quad c_0 \neq 0$$

~~$y_1'(x) =$~~

$$y_1(x) = x^{r_2} \sum_{n=0}^{\infty} c_n x^n$$

$y'' - 4y' + 4y = 0$

$r^2 - 4r + 4 = 0$

$r = 2$

$y_H = c_1 e^{2x} + c_2 x e^{2x}$

$y_{özel}$

$y_{özel} = A\sin 2x + B\cos 2x$

$y' = 2A\cos 2x - 2B\sin 2x$

$y'' = -4A\sin 2x - 4B\cos 2x$

$= -4A\sin 2x - 4B\cos 2x$

$-4A\sin 2x - 4B\cos 2x - 8A\cos 2x + 8B\sin 2x + 4A\sin 2x + 4B\cos 2x = \sin 2x$

$8B = 1$

$y_H = c_1 e^{2x} + c_2 x e^{2x} - \frac{1}{8}\sin 2x$

$B = -\frac{1}{8}$

$y_{özel} = -\frac{1}{8}\cos 2x \cdot \frac{\pi}{8}$

$y_{özel} = -\frac{1}{8}\sin 2x \cdot \frac{\pi}{4}$

$= -\frac{1}{8}\cdot\frac{\sqrt{2}}{2}$

$y_{özel} = A\sin 2x + B\cos 2x$

$y' = 2A\cos 2x$

$y = \sum_{n=0}^{\infty} a_n x^n$

$\sum_{n=0}^{\infty} n a_n x^{n-1} = 0 + \sum_{n=1}^{\infty} n a_n x^{n-1}$

$y' = \sum_{n=1}^{\infty} n \cdot a_n x^{n-1}$

$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

denklemde yerine yazalım

$$(x+1)\left(\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}\right) - \left(\sum_{n=0}^{\infty} a_n x^n\right) = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-1} + \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^n = 0$$

↳ x'li ifadeler

$$\sum_{n=1}^{\infty} (n+1) n\, a_{n+1} x^n + \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=0}^{\infty} a_n x^n = 0$$

↳ n'ler aynı olsun

$$2a_2 - a_0 + \sum_{n=1}^{\infty} \left[(n+1) n\, a_{n+1} + (n+2)(n+1) a_{n+2} - a_n\right] x^n = 0$$

$2a_2 - a_0 = 0$

$\boxed{a_0 = 2a_2}$

$n = 1$ için

$2a_2 + 3 \cdot 2 a_3 - a_1 = 0$

$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x^1 + a_2 x^2 + a_3 x^3$

$y(0) = 0 \quad a_0 = 0 \Rightarrow 2a_2 = 0 \Rightarrow a_2 = 0$

$y'(x) = a_1 + 2a_2 x + 3a_3 x^2$

$y'(0) = 1 \qquad a_1 = 1$

$0 + 6a_3 - 1 = 0$

$a_3 = +\frac{1}{6}$